

A disproof of Conjecture 00000008185

$G(k) \neq k$ for every $k \geq 2$, so “ $G(k) = k$ ” has no threshold

1 The conjecture

Conjecture 00000008185 reads (English version):

Definition: The minimal bases of power-sum representations of integers: the computational threshold $\text{waring_c}(k)$ of $g(k)$ and $G(k)$ of Waring’s problem. Conjecture: The optimal upper bound $G(k) \leq k(\log k + \log \log k + C)$ holds with $C = 4$ an explicit tightening of the Vaughan-type constant and a remainder $O(k/\log k)$ (a logarithmic remainder law); the exceptional set of the complete formula for $g(k)$ (the 157 exceptions at $k = 4$) is finite (a finite exception law), the last element being 79 (a seventy-nine last-exception rigidity); and the high-dimensional generalization of the known rigidity $G(4) = 16$ – namely $G(k) = k$ – first holds at threshold k at least 7 (a seventh-power asymptotic threshold).

The Chinese version of the last clause reads, in transliteration, “ $G(4)=16$ de shiliu gangxing (yizhi) de gaoweituguang $G(k)=k$ de zuixiaok weik ≥ 7 de jianjin chengliyu (qici jianjin yu)”: “the minimal k for the higher-dimensional generalization $G(k) = k$ of the (known) rigidity $G(4) = 16$ is the asymptotic threshold $k \geq 7$ ”.

The conjecture is a conjunction of three laws. **Result.** The third law (the seventh-power threshold) is false under each reading listed in Section 2, so the conjunction is false. The reason is the classical lower bound $G(k) \geq k + 1$ for $k \geq 2$, which we reprove from scratch by counting; in particular $G(k) = k$ holds for no $k \geq 2$. Everything is formalized in Lean 4 with Mathlib (Section 4).

2 Definitions, readings and conventions

Definition 1 (Hardy–Littlewood $G(k)$). $G(k)$ is the least positive integer s such that every sufficiently large integer is a sum of at most s positive k -th powers. Source: Wikipedia, *Waring’s problem* (https://en.wikipedia.org/wiki/Waring%27s_problem): “ $G(k)$ is defined to be the least positive integer s such that every sufficiently large integer (i.e. every integer greater than some constant) can be represented as a sum of at most s positive integers to the power of k .”

In Lean: `IsSumAtMost k s n` says $n = a_1^k + \dots + a_m^k$ with $m \leq s$ and all $a_i \geq 1$ (a list of positive integers); `Suffices k s` says $\exists N \forall n \geq N, \text{IsSumAtMost } k \ s \ n$; and `waringG k` is `sInf {s | Suffices k s}` over $\mathbb{N} = \{0, 1, 2, \dots\}$.

Conventions. (i) $s = 0$ never suffices (large n is not an empty sum), so the infimum is over positive s , as in the definition. (ii) Hilbert’s theorem that some s suffices is not in Mathlib. If no s sufficed, Lean’s `sInf` of the empty set would be 0; the results below do not depend on this junk

value: $G(k) \neq k$ holds either way, and $G(k) \geq k + 1$ is stated under the hypothesis that some s suffices. (iii) Sums of at most s positive k -th powers are the same as sums of exactly s nonnegative k -th powers (pad with zeros); the proof uses the latter for counting.

Readings of the last clause refuted.

(R1) $G(7) = 7$, or $G(k) = k$ for all $k \geq 7$.

(R2) There is a threshold $K \geq 7$ with $G(k) = k$ for all $k \geq K$ (“asymptotic threshold”).

(R3) The least $k \geq 2$ with $G(k) = k$ exists and is at least 7.

(R4) The least $k \geq 1$ with $G(k) = k$ is at least 7 (false since $G(1) = 1$; Wikipedia: “Clearly, $G(1) = 1$ ”).

3 Proof

Lemma 2 (padding and counting). *Let $k \geq 1$ and $t \geq 0$. If $n \leq t^k$ is a sum of at most k positive k -th powers, then $n = \sum_{b \in m} b^k$ for a multiset m of exactly k elements of $\{0, 1, \dots, t\}$. There are $\binom{t+k}{k}$ such multisets, so at most $\binom{t+k}{k}$ integers $n \leq t^k$ are such sums.*

Proof. Each base a satisfies $a^k \leq n \leq t^k$, so $a \leq t$. Pad the list of bases with zeros to length k ; since $0^k = 0$ the sum is unchanged. The number of k -multisets from a set of $t + 1$ elements is $\binom{t+1+k-1}{k} = \binom{t+k}{k}$ (Mathlib: `Sym.card_sym_eq_choose`). $\square$

Lemma 3. *For $k \geq 2$ and every $M \geq 0$, with $t = 3k + 4^k M$ and $C = \binom{t+k}{k}$, we have $C + M \leq t^k$.*

Proof. (a) $k!C = (t+k)(t+k-1)\cdots(t+1) \leq (t+k)^k$. (b) $3(t+k) \leq 4t$ since $t \geq 3k$, so $3^k(t+k)^k \leq 4^k t^k$. (c) $4^k + 1 \leq 3^k k!$ for $k \geq 2$: for $k = 2$, $17 \leq 18$; and $4^{k+1} + 1 \leq 4(4^k + 1) \leq 4 \cdot 3^k k! \leq 3^{k+1}(k+1)!$ since $4 \leq 3(k+1)$. (d) $C = \binom{t+k}{k} \geq \binom{t+1}{t} = t + 1 > 4^k M$. From (a)–(c), $(4^k + 1)C \leq 3^k k! C \leq 3^k(t+k)^k \leq 4^k t^k$. With (d), $4^k(C + M) \leq 4^k C + C \leq 4^k t^k$; divide by 4^k . $\square$

Theorem 4. *For every $k \geq 2$ and every M , some $n \geq M$ is not a sum of at most k positive k -th powers. Hence no $s \leq k$ suffices for all sufficiently large integers.*

Proof. Take t as in Lemma 3. The interval $[M, t^k]$ contains $t^k + 1 - M > C$ integers, and by Lemma 2 at most C of them are sums of at most k positive k -th powers. A sum of at most $s \leq k$ powers is also a sum of at most k powers. $\square$

Corollary 5. *For every $k \geq 2$: $G(k) \neq k$; and $G(k) \geq k + 1$ whenever some s suffices. Also $G(1) = 1$. Consequently (R1)–(R4) are all false.*

Proof. If $G(k) = k \geq 2$, then the set $\{s : s \text{ suffices}\}$ is nonempty (otherwise its infimum would be 0), so its infimum k belongs to it, contradicting Theorem 4. If the set is nonempty, its least element is not $\leq k$, so it is $\geq k + 1$. For $k = 1$: $n = n^1$ shows $s = 1$ suffices, and $s = 0$ does not. (R1) fails at $k = 7$; (R2) fails at $k = K$; (R3) fails because no $k \geq 2$ has $G(k) = k$; (R4) fails because $G(1) = 1$ forces the least such k to be $1 < 7$. $\square$

4 Lean formalization

File `lean/Conjecture8185/Basic.lean` (179 lines, namespace `C8185`; only import `Mathlib`). Definitions: `IsSumAtMost`, `Suffices`, `waringG`, `toFin`, `powSum` (sum of k -th powers over a `Sym (Fin (t+1)) k`). Lemmas: `mem_image_powSum` (Lemma 2); `four_pow_lt`, `choose_add_le` (Lemma 3); `exists_not_sum`, `not_suffices` (Theorem 4); `waringG_ne`, `waringG_ge`, `waringG_one` (Corollary 5). Main theorem (ASCII rendering):

```
theorem conjecture_8185_false :
  (forall k, 2 <= k -> forall N, exists n >= N, not IsSumAtMost k k n) and
  (forall k, 2 <= k -> waringG k != k) and waringG 1 = 1 and
  waringG 7 != 7 and
  not (forall k, 7 <= k -> waringG k = k) and
  not (exists K, 7 <= K and forall k, K <= k -> waringG k = k) and
  not (exists k, 2 <= k and waringG k = k) and
  not (exists k0, 7 <= k0 and waringG k0 = k0 and
    forall k, 1 <= k -> waringG k = k -> k0 <= k)
```

Axiom audit. `#print axioms C8185.conjecture_8185_false` reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`. The file compiles under `lake build` and with `-DwarningAsError=true`.

5 Scope

- The clause is refuted literally, with $G(k) = k$ as written in both languages. If “ $G(k) = k$ ” were a garbling of another generalization of $G(4) = 16$ (for instance $G(k) = 4k$, k^2 , 2^k or $k + 1$), nothing here refutes it.
- The first law ($G(k) \leq k(\log k + \log \log k + 4)$ with a remainder) is of Wooley type and is not addressed. The second law (“157 exceptions at $k = 4$ ”, “last element 79”) is not addressed either. Wikipedia notes that “79 requires 19 fourth powers”.
- The witness exponents are not degenerate: the failure holds for every $k \geq 2$, including the stated threshold $k = 7$. Only reading (R4) also uses $k = 1$.